Introduction to Algorithm Engineering
Quiz Examination

March 31, 2026

Instructions:

- The quiz is for 45 minutes and 20 points.
- There are four questions in Section I and two questions in Section II. Make sure your question paper is printed correctly.
- No clarifications shall be provided during the exam. If you need to make any assumptions, state those explicitly.

Section I: Answer the following questions in brief.

1. Define the terms biconnected components and strongly connected compnents of an undirected graph and a directed graph, respectively. **(Points:1+1=2)**
2. What is meant by the inclusive property of a cache? Explain with an example. **(Points:1+1=2)**
3. Find the articulation points of the graph in Figure 1 using any of the algorithms discussed in class.

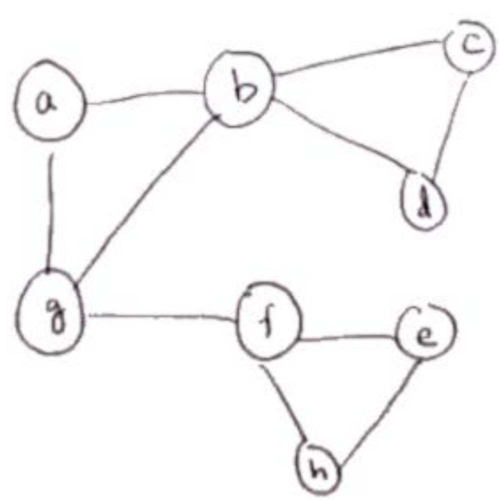

Figure 1: Graph for Question 3, Section A

(Points:2)

4. Consider the mesh graph of $n \times n$ vertices as shown in Figure 2. If we use the algorithm of Crescenzi to find the diameter of this graph, how many BFS traversals are needed? Justify your answer. **(Points:2)**

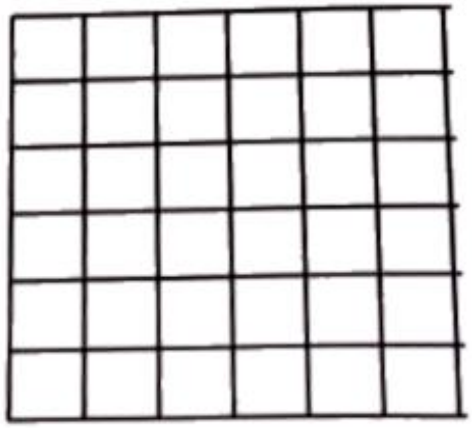

Figure 2: A mesh graph of size $n \times n$.

Section II: Answer each question with enough details.

1. Recall the problem of transposing a square matrix of dimensions $n \times n$. Answer the following questions.

    (a) Write a simple algorithm for the transpose of a matrix and estimate the number of cache misses occured.

    (b) Write a cache-aware algorithm to transpose a square matrix.

    (c) Estimate the number of cache misses incurred.

    (Points:1+2+3=6)

2. Answer the following questions with respect to strongly connected components.

    (a) Suppose a strongly connected component C of a directed graph G has k vetices. What is the smallest and largest number of edges with each edge having both its endpoints in the component C?

    (b) How does the answer to the above question change if we require that only one end point is in the component and the other is not in the component C?

    (c) Suppose that each recursive call in the algorithm we studied in class to find the strongly connected components of a directed graph actually results in equal sized subproblems. What is the run time of the algorithm under this assumption. Justify your answer.

    (Points:2+2+2=6)